

- 6 (a) State what is meant by diffraction.

.....
 [1]

- (b) Light of wavelength 720 nm in a vacuum is incident normally on a diffraction grating as shown in Fig. 6.1.

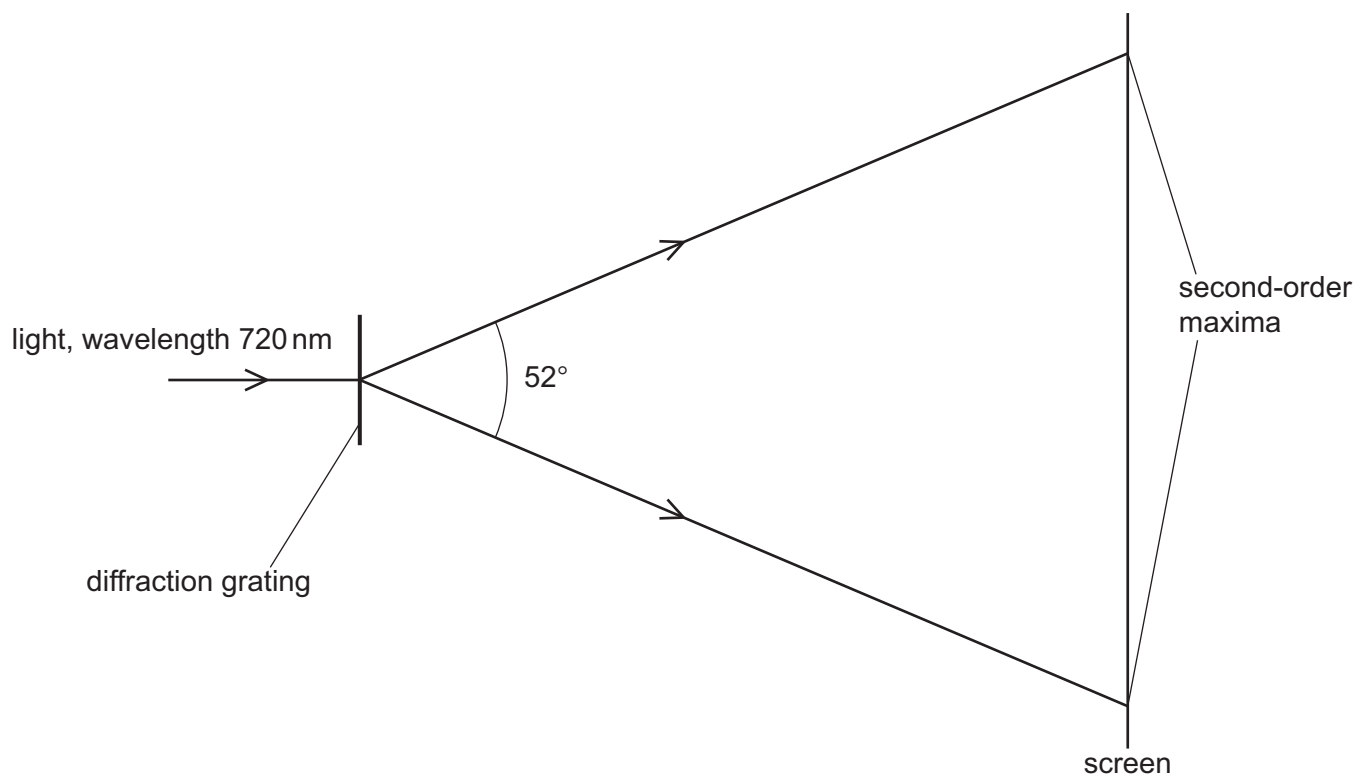


Fig. 6.1 (not to scale)

A screen is parallel to the grating. An interference pattern is seen on the screen and the angle between the **second-order** maxima is 52° .

- (i) Calculate the frequency of the light.

frequency = Hz [2]



- (ii) Calculate the number of lines per unit length in the diffraction grating.

number per unit length = m^{-1} [3]

- (iii) The light in Fig. 6.1 is now replaced with light of a different wavelength λ . It is observed that the third-order maxima of this light are at the same positions as the second-order maxima of the light in Fig. 6.1.

Calculate, in nm, the wavelength λ .

$\lambda =$ nm [2]

[Total: 8]

